

Artificial Intelligence and Retrieval-Augmented Generation

1. Introduction to RAG

Retrieval-Augmented Generation, or RAG, is an AI architecture that enhances large language models by retrieving relevant facts from an external knowledge base before generating a response. This reduces hallucinations significantly.

2. Core Components of Local RAG

- Document Processing: Loading files like PDFs and chunking text into pieces.
- Embeddings: Converting text into high-dimensional numerical vectors.
- Vector Database: FAISS is used to perform fast similarity search.
- Local LLM: Models like Qwen generate accurate answers based on context.

3. Advantages of Running RAG Locally

Running RAG locally provides 100% data privacy since no data is sent to external APIs. It works completely offline, without API subscription fees or rate limits.